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PALEOMAGNETISM OF LATE CRETACEOUS AND EOCENE LIMESTONES AND CHALKS FROM HAITI: TECTONIC INTERPRETATIONS

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Abstract. We have conducted a paleomagnetic study of limestones and chalks from Haiti which suggests that the island of Hispaniola is a composite of at least two tectonically independent blocks. Sampling sites are distributed among three widely separated localities, namely: Les Cayes (nine sites; 18.3° N. Latitude; 73.6° W. Longitude) and Beloc (five sites; 18.4° N. Latitude; 72.6° W. Longitude) which are located on the southern peninsula of Haiti, and a northern locality, Ennery (six sites; 19.5° N. Latitude; 72.5° W. Longitude). Stable magnetization components of dual polarity and moderately high blocking temperature were revealed through progressive thermal demagnetization of samples from all of the Beloc and Ennery sites, while seven of the nine Les Cayes sites possessed unstable magnetizations. Mean paleolatitudes calculated from Beloc and Ennery data show 8° of latitudinal separation that is of at least Eocene age. Pole positions (Beloc: 43.2° N. Latitude; 185.8° E. Longitude, $\alpha 95$ = 25.1° and Ennery: 66.1° N. Latitude; 31.3° E. Longitude, $\alpha 95$ = 19.7°) show both colatitudinal and angular discordance with the North American apparent polar wander path. Based on these disparities, we demonstrate that the implied tectonic displacements tend to support some aspects of previously published Caribbean plate tectonic models.

INTRODUCTION

Published Caribbean plate tectonic models are varied and numerous. Some are based on geologic extrapolation [e.g. Malfait and Dinkelman, 1972; Anderson and Schmidt, 1983; Pindell and Dewey, 1982; Burke et al., 1984; Duncan and Hargraves, 1984] and others emphasize recent seismicity [Molnar and Sykes, 1969; Sykes et al., 1982]. In general, there are two aspects which are agreed upon in the literature. Convergence between the Caribbean and North American plates, during which time the Greater Antilles island arc terrain was formed, expired around Eocene time with the collision of the Greater Antilles and Bahamas platform. Since then, eastward migration of the Caribbean plate relative to North and South America has subducted central Atlantic lithosphere giving rise to volcanism in the Lesser Antilles.

The paleogeography of the Greater Antilles in the Late Cretaceous and Early Tertiary is still poorly constrained, as emphasized by the contrasting reconstructions which have been recently proposed [c.f. Pindell and Dewey, 1982; Anderson and Schmidt, 1983]. The past relationship between the southern peninsula of Haiti and the rest of Hispaniola is also unclear [Sykes et al., 1982; Mann et al., 1984]. Rotational displacements of lithospheric blocks can be quantitatively constrained by paleomagnetism, and the style and distribution of strike-slip-induced rotations at the northern Caribbean plate boundary may help to differentiate between the various rotational mechanisms for crustal blocks in strike-slip zones.

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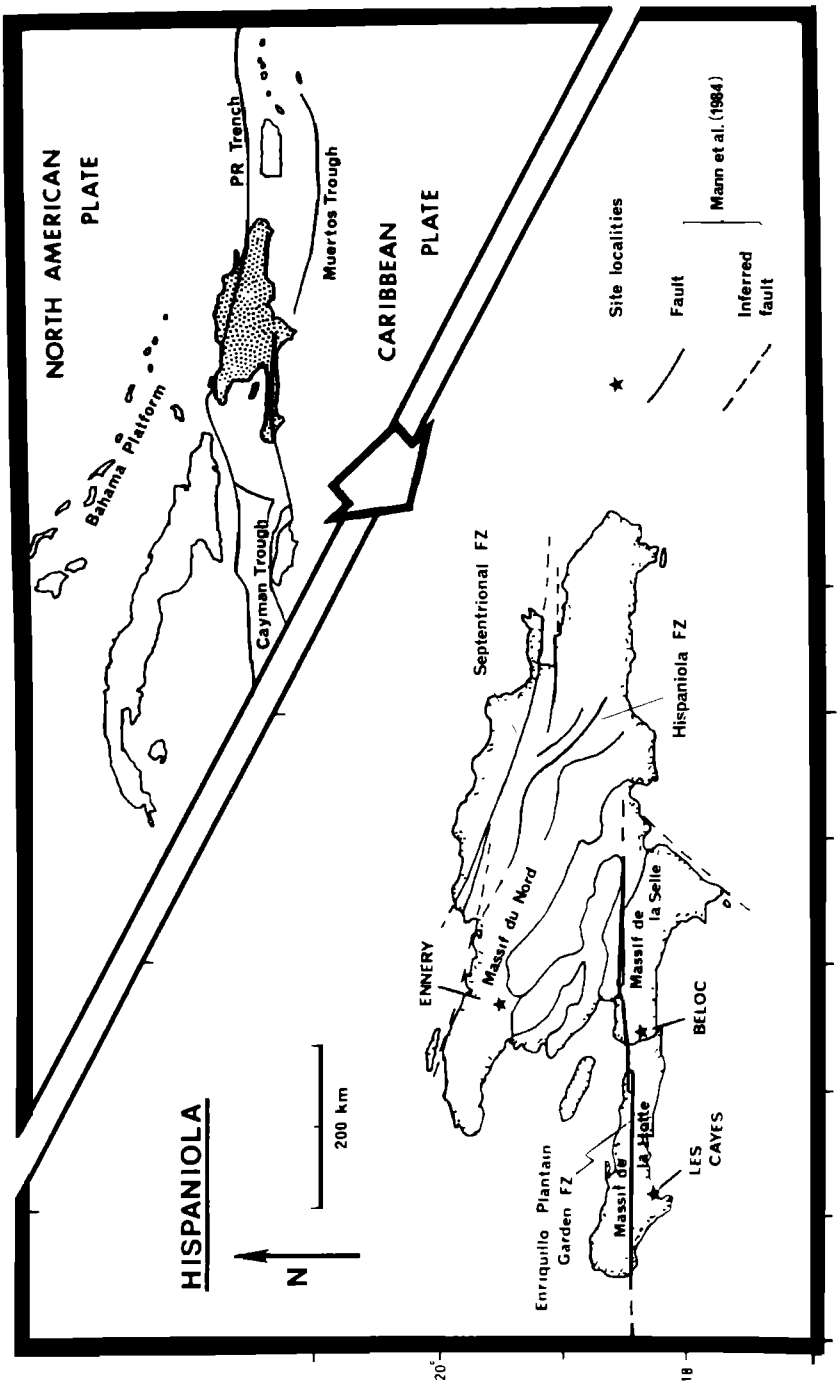


Fig. 1. Sketch maps of the Greater Antilles and Hispaniola showing the three sampling localities of this study. The location of major faults on the island are from Mann et al. [1984]: Note especially the Enriquillo-Plantain Garden Fault Zone and the Hispaniola Fault Zone which are discussed in the text. Also shown are some of the major physiographic provinces of Hispaniola.

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TABLE 1. Site Mean Data From Haiti

Site	Rs	Bedding Dip/Az	Uncorrected Dec.	Uncorrected Inc.	Corrected Dec.	Corrected Inc.	α95	k
			Beloc (18.4 N. Latitude, 072.6 W. Longitude)					
H08	55/69	18/068	302.9	0.3	303.7	9.6	7.6	7.3
H09*	3/6	15/070	88.1	11.4	87.4	-1.0	53.2	6.4
H10	7/7	10/115	332.6	23.8	335.8	28.5	8.7	49.2
H11	5/7	12/097	119.6	14.1	118.9	2.1	24.6	14.9
H12	4/5	25/123	143.3	16.9	141.9	-0.5	14.8	39.5
			Ennery (19.5 N. Latitude, 072.5 W. Longitude)					
H13	5/6	14/240	203.2	-5.5	201.7	-17.6	13.3	33.8
H14	6/6	26/250	11.7	36.4	350.0	49.0	11.7	33.7
H15	7/9	09/220	10.4	10.5	9.8	17.6	18.9	11.1
H16	4/8	10/220	215.4	-23.7	215.7	-33.7	36.9	7.2
H17	3/7	08/125	214.8	-1.8	215.0	-1.0	32.3	15.7
H18	7/8	07/200	218.7	-12.4	219.6	-18.8	16.7	14.0
			Les Cayes (18.3 N. Latitude, 073.6 W. Longitude)					
H02	7/9	19/250	19.3	24.5	12.7	31.8	45.3	2.7
H03	4/10	22/225	65.3	-3.5	66.7	15.5	53.2	4.0

The Beloc and Ennery data are plotted in Figure 4. Rs, ratio of number of samples used in computation to number of samples measured; bedding dip/az, average of bedding attitudes measured at the site; dec., inc., declination, inclination; α95, radius in degrees of the cone of 95% confidence; k, estimate of the precision parameter K. Note (*) site H09 eliminated from pole calculation due to high α95 value.

Paleomagnetic data from Hispaniola are very scarce. Vincenz and Dasgupta [1978] obtained paleomagnetic data from three igneous sites in the Dominican Republic, ranging in age from Cretaceous to Miocene. The data were interpreted in terms of large anticlockwise rotations of Hispaniola since the Cretaceous relative to North America. Kent and Maurrasse [1982] sampled the basalts of the Dumisseau Formation in southern Haiti and considered them totally remagnetized. In this paper we do not attempt to formulate a new tectonic model; rather, the objective of our study is to provide a new and independent paleomagnetic data set by which various aspects of proposed models may be tested.

SAMPLING LOCALITIES

We have collected samples from three widely separated localities in Haiti. These are designated from north to south as Ennery, Beloc, and Les Cayes [Figure 1]. A "locality" consists of several sites, each of which in turn represents 7-11 cored samples. Sites within a locality are distributed about an area of a few square kilometers; a distribution that will become important in our final synthesis and interpretation of the paleomagnetic data.

Samples were obtained in the field using a portable rock coring drill. At the Beloc locality we used the oriented-block method to collect the less consolidated chalk lithologies. All orientations were made using the Brunton compass, and at each site we were careful to sample only the freshest appearing materials. Field information regarding the number of sites and number of samples at each locality is included in Table 1.

Ennery

Eocene age carbonates are widespread in northwest Hispaniola. Along the northwest-southeast trending Massif du Nord (Figure 1) middle and upper Eocene limestones attain thicknesses of approximately 1 km. We have sampled the Plaisance Formation at six separate exposures approximately 20.9 km northeast of Gonaives, near the town of Ennery. The bedded limestones here have a shallow dip (about 10°) directed toward the southwest and are generally of a pale yellow to light tan color. At one site (H17) the bedding direction switches to the

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southeast, which may indicate a regional fold in the area. Woodring [1922] assigned a middle Eocene age to these limestones recognizing both the existence of several index species of Dictyoconus and the absence of large orbitoidal foraminifera, characteristic of upper Eocene limestones in Haiti.

Beloc

The recently named Beloc Formation has been described in detail by Maurrasse [1982]. These pelagic chalks and limestones comprise the western end of the Massif de la Selle region of Hispaniola, and our sampling area is located 3.8 km south of Beloc at the type section (see Figure 1). Beloc carbonates rest unconformably above basalts of the Dumisseau Formation which is considered the ancient analog of the nearby Caribbean seafloor. The Beloc Formation attains a total thickness of about 150 m which spans the Late Cretaceous Globotruncana contusa Zone to perhaps the Globorotalia pseudobulloides Zone of the earliest Paleocene [Maurrasse, 1982]. One of our five sampling sites at Beloc comprises a continuous stratigraphic section through the Cretaceous-Tertiary boundary. Although biostratigraphic and magnetostratigraphic study of this section is still in progress, our preliminary findings have significant implications regarding the tectonic interpretation of the paleomagnetic data from Haiti.

Les Cayes

Further west along the southern peninsula we have sampled nine outcrops of pure white to light gray, cherty limestones of the Massif de la Hotte region. The rocks are exposed along the highway west from Jacmel, about 15 km east of Les Cayes (Figure 1). At several exposures we observed siliceous, dark gray limestone beds of about 5 cm thickness interbedded with the white limestone. The rocks here are gently folded, but occasionally we observed very sharp changes in bedding dips. Previous descriptions of these rocks are difficult to find in the literature. However, Woodring et al. [1924] described upper Eocene limestones throughout the Massif de la Hotte and in the same work, determined a late Eocene age for light to dark gray limestones about 15 km east of Les Cayes (see Figure 1). In this study the limestones sampled east of Les Cayes are assumed to be of late Eocene age.

PALEOMAGNETISM

All samples were measured with an A100 series United Scientific superconducting rock magnetometer having sensitivity to magnetic moment of 10^{-5} Am^{-1}. Progressive thermal and alternating field (AF) demagnetizations were conducted using Schonstedt models TSD-1 thermal and GSD-1 single axis AF demagnetizers, respectively. Two Mu-metal shielding compartments were used to store samples before and during analyses.

Progressive Demagnetization

The mean natural remanent magnetization (NRM) intensity at Ennery is 1.71 $x10^{-4}$ Am^{-1}, at Les Cayes the mean is 8.38 $x10^{-4}$ Am^{-1}. The strongest NRM intensities were measured in the limestones and chalks at the Beloc locality, where the mean is 6.42 $x10^{-3}$ Am^{-1} and intensity range is 2.98 $x10^{-4}$ to 1.18 $x10^{-2}$ Am^{-1}. Approximately 20% of the samples were subjected to a detailed preliminary demagnetization study. This pilot suite of samples contained representatives from all 20 sites and all lithologies present in the collection . In this manner, optimum demagnetization schemes for various sites/lithologies can be determined and utilized in subsequent analyses. All demagnetization work is expressed graphically using orthogonal vector diagrams [Zijderveld, 1967] through which components of magnetization can be visually isolated. Our data were subjected to computerized principal component analysis according to the method of Kirschvink [1980]. Fisher [1953] statistics were used to estimate the error in calculation of site mean directions.

Ennery. These samples, having among the weakest NRM intensities, yielded stable components of remanence. Representative orthogonal demagnetization diagrams from the Ennery locality are shown in Figure 2. Nearly all of the samples were found to contain a moderately high blocking temperature, bipolar component of magnetization (north by northeast, positively inclined or south by southwest, negatively inclined). This characteristic component was resolved within temperatures of typically 250°-355°C for sites H15, H16 and H18; 175°-350°C for site H14; and 300°-430°C for site H13.

The magnetizations decay on an origin-bound trajectory, but often near the final demagnetization steps, remanence intensities approach the sensitivity of the cryogenic magnetometer and diverge from the origin (Figures 2a and 2d). It is presumed that at these steps the measurements are those of background noise that has dominated the very weak magnetic moment intensities remaining in these samples.

Beloc. Progressive thermal demagnetization of the Beloc chalks and limestones (Figures 3a-3f) revealed very stable components of remanence in four of the five sites. Characteristic, high blocking temperature components, resolved in a range of 500°-580°C from sites H08 and H10-H12, are antipodal (northwest and positive or southeast and negatively inclined). This direction is very different from the Ennery characteristic directions. Results from site H09 were

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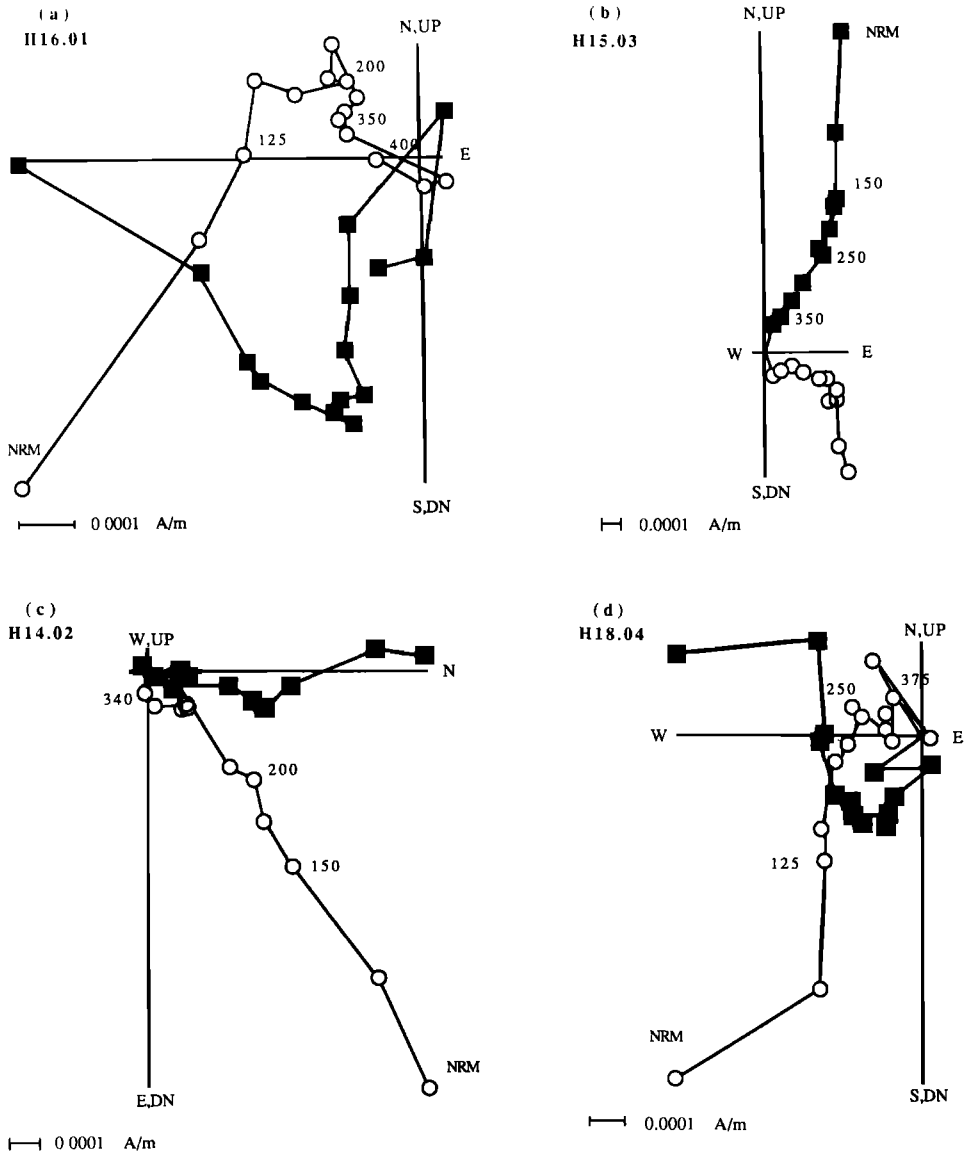


Fig. 2. (a-d) Orthogonal demagnetization diagrams representing progressive thermal demagnetization of samples from the Ennery locality. In each diagram an open (closed) symbol represents the vertical (horizontal) projections of the endpoint of the magnetization vector plotted in bedding corrected coordinates. Sample name, temperature range (in °C), and magnetization intensity scale are provided in each diagram. Note that in Figure 2c, for clarity, the demagnetization record of sample H14.02 is plotted with up-down versus N-S.

ambiguous; most of the magnetizations were unstable, with two relatively "stable" magnetizations lacking directional consistency.

Diagrams shown in Figures 3d-3e typify the response of Beloc carbonate to thermal cleaning. A low blocking temperature component (northward and down) is removed at about 250°C. From this temperature the magnetization decays toward the origin in fairly linear fashion. The blocking temperature spectra of these two components overlap slightly in the range of 250°-300°C as demonstrated by the curvilinear demagnetization paths. However, the most remarkable characteristic of these diagrams is the very shallow inclinations we observe. At this juncture in our study we noted that if these bedding-corrected, high blocking temperature components indeed represented primary magnetizations, then these chalks and limestones were deposited very near

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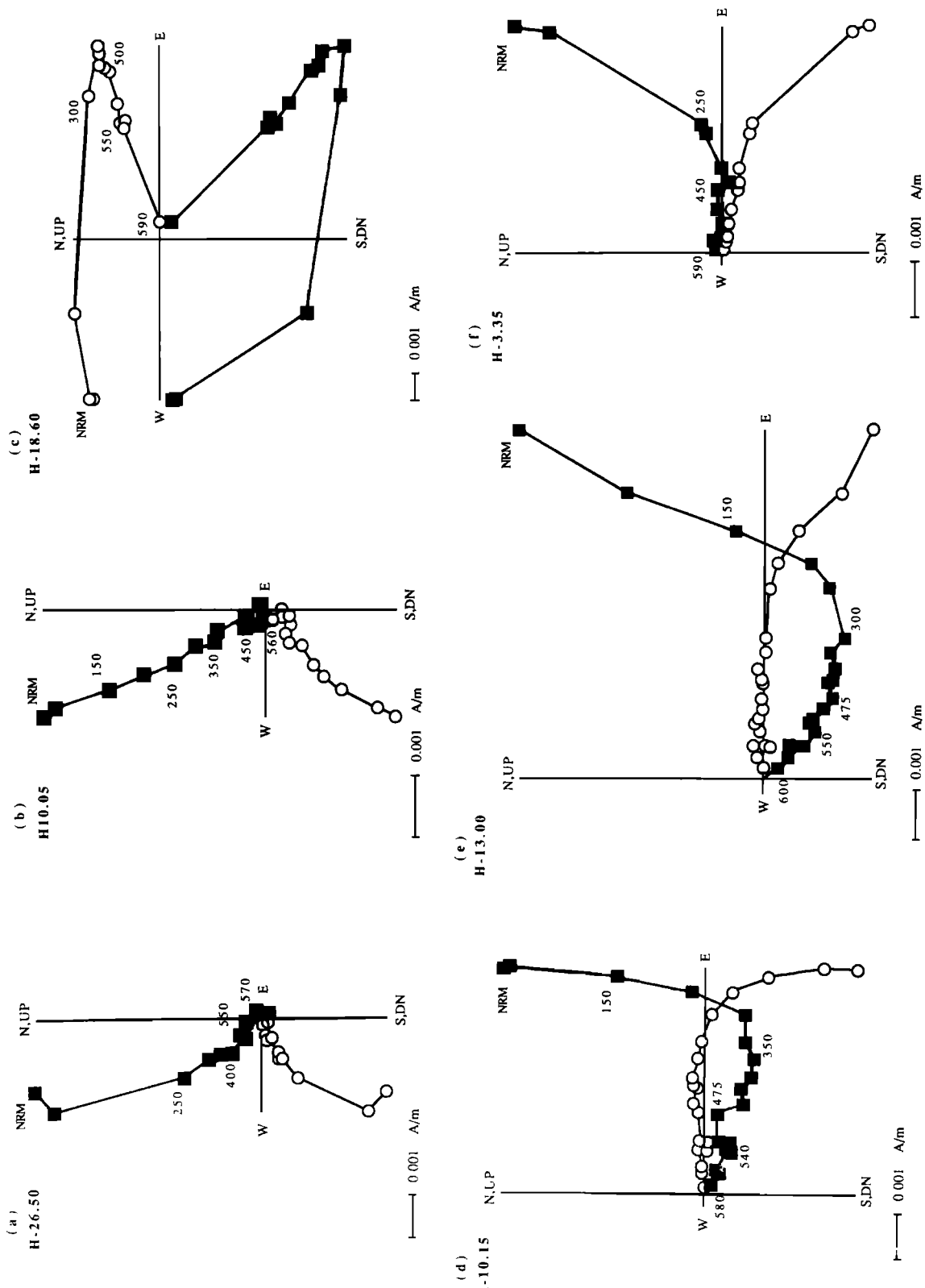


Fig. 3. (a-f) Orthogonal demagnetization diagrams representing progressive thermal demagnetization of Beloc samples. Open (closed) symbols represent the vertical (horizontal) projection of the endpoint of the magnetization vector plotted in bedding corrected coordinates. Sample name, temperature range (in °C), and magnetization intensity scale are provided in each diagram.

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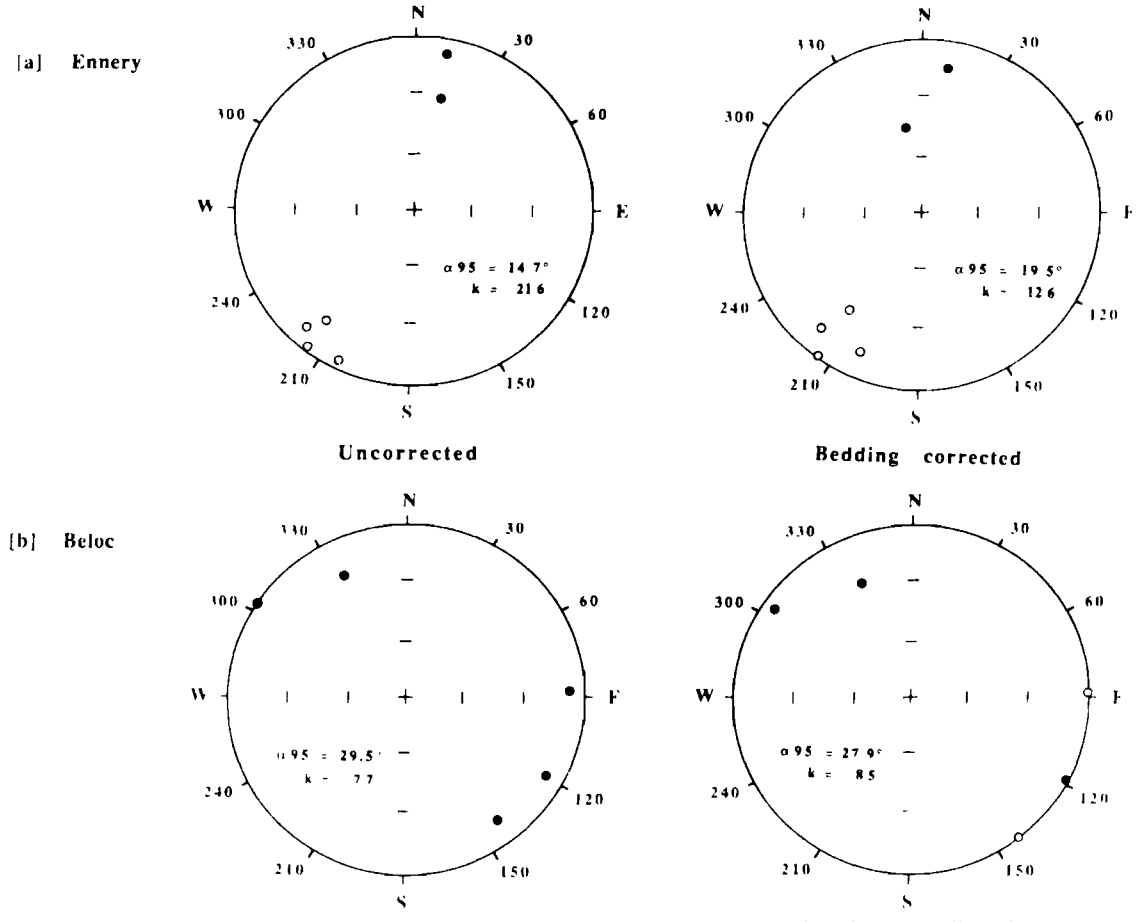


Fig. 4. (a) Site mean data from the Ennery locality in northern Haiti. Corresponding data are listed in Table 1. Note the dual polarity, the north by northeast, south by southwest trend, and the slight increase in dispersion upon application of bedding corrections. Open (closed) symbols fall in the upper (lower) hemisphere. (b) Site mean data from the Beloc locality, southern Haiti. Corresponding data are listed in Table 1. Dispersion of these data decreases slightly upon application of bedding corrections. The site means show a dual polarity and follow a northwest-southeast trend.

the paleoequator. Given these very shallow inclinations, we also noted that dispersion in our data would make "polarity" assignments based solely on the vertical projection very difficult. However, nearly all of the characteristic directions fell into two groups: one northerly and positive, the other southerly and negative. The orthogonal projection shown in Figure 3f, however, is one case in which an easterly component displayed a slightly positive inclination.

Les Cayes. Progressive thermal demagnetization yielded stable components in a few samples from only two of the nine Les Cayes sites (sites H02 and H03). An unstable, erratic response to thermal cleaning can be observed in all of the remaining sites. Although sites H02 and H03 display a stable component unblocked at moderately high temperatures (200°-390°C), no consistent magnetization direction is revealed through vector analysis of this component among the samples from these two sites. We further analyzed these thermally unstable rocks using the alternating field method of cleaning. The typical response of the limestones to applied AF fields is as unsatisfactory as that obtained through thermal cleaning attempts.

Site Mean Directions

Principal component directions obtained through least squares analysis [Kirschvink, 1980] of stable magnetizations were used in calculation of site means. These data are listed in Table 1 along with corresponding Fisher statistics. The fact that any given site mean assumes a particular "polarity" is

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TABLE 2. Summary of Pole Position Data and Comparison to the NAAPW Path of Irving and Irving [1982]

Locality	N	Observed Dec.	Observed Inc.	α95	Pole N.Lat.	Pole E.Long.	dθ	dm	Ideal Dec.	Ideal Inc.	Reference Pole	Δ
Ennery	6	023.7	23.6	19.7	66.1	031.3	11.2	21.0	347	32	50Ma	+37°
Beloc	4	314.3	8.8	25.1	43.2	185.8	12.8	25.3	347	30	60Ma	-33°

Poles are plotted in Figure 5. Dec., inc., declination, inclination; n. lat., e. long., north latitude, east longitude. Variables dθ and dm are, respectively, the radii of the ellipse of 95% confidence along, and perpendicular to the site-to-pole magnetic meridian. Ideal dec. and inc. refers to the paleomagnetic directions expected of data from each sampling locality if no rotations occurred relative to the appropriate segment of the NAAPW path [i.e. the reference pole]. The Δ value refers to the horizontal angular discordance between the ideal and observed directions measured about a vertical axis positioned at the sampling locality. Positive (negative) Δ value indicates a clockwise (anticlockwise) rotation.

simply an artifact of a computer algorithm which, in order to calculate a site mean, converts data to a common hemisphere.

For each locality, site mean directions in both geographic and tilt-corrected coordinates are shown in the stereographic projections of Figure 4. From these figures and Table 1, bipolarity among the sites can be observed. Site means H16 and H17 have relatively high α95 values (36.9° and 32.3°, respectively) yet they are consistent in direction with other Ennery site means. Their high scatter is partly the result of the small number of samples used in the computation (see Table 1). The overall Ennery site mean trend is very different from the northwest-southeast tendency of Beloc data (Figure 4a). One site mean (H09) is displaced from this trend. It has the highest α95 value (53.2°) and is calculated using only three principal component directions.

As shown in Figure 4a, grouping of the Beloc site means improves slightly upon application of the tilt correction. The opposite is true in the case of the Ennery data (Figure 4b), but in both instances, site mean dispersion changes are insignificant at 95% level of confidence. Obviously, if fold tests were possible at these localities, a more conclusive statement could be made regarding the timing of remanence acquisition. In the absence of a proper field test, directional consistency and bipolarity among the Beloc and Ennery site means argues for a primary remanence.

POLE POSITIONS AND TECTONIC IMPLICATIONS

Before we calculate pole positions from our data we must decide which of the antiparallel magnetization components is representative of the north seeking paleomagnetic direction. In the case of the Caribbean this decision is not always straightforward due to the low inclination values associated with Mesozoic and Cenozoic paleomagnetic data, and the poor constraints on paleogeographic models which could place the Greater Antilles in either hemisphere during the Late Cretaceous. However, our preliminary study of geomagnetic polarity stratigraphy at Beloc (site H08) in combination with the biostratigraphy of Maurrasse [1982] provides us with the data necessary to distinguish the two polarities. Polarity chron 29R spans the Cretaceous-Tertiary boundary and the corresponding reversed magnetozone has been identified in many pelagic boundary sections [e.g. Alvarez et al., 1977 and Channell and Medizza, 1981]. Maurasse [1982] has identified the Cretaceous—Tertiary boundary within the chalk and limestone stratigraphy at Beloc. Twenty samples at site H08 cover the stratigraphy across this boundary and all contain a characteristic magnetization with southeasterly declination and shallow negative inclination. As the Earth's magnetic field was reversed in polarity during the Mesozoic-Cenozoic transition, these southeasterly directions represent reversed magnetizations, and the negative inclinations demonstrate that Hispaniola was in the northern hemisphere during the latest Cretaceous.

We now compute pole positions from the Beloc data using the northwest and positively inclined orientation, a necessary condition when comparing derived poles with a North American apparent polar wander (NAAPW) path composed of north poles. We have extended this reasoning to the Ennery data, considering that this locality is today, north of Beloc and probably was so in the past. Locality mean and pole position data are summarized in Table 2. The two sites from Les Cayes were both judged as unreliable for further analysis due to the high α95 values (Table 1). Data from site H09 have been eliminated from the Beloc pole calculation (α95 = 53.2°), and therefore N = 4 for this pole. Locality mean data for Beloc and Ennery (α95 = 25.1° and 19.7°, respectively, see Table 2) are offset by approximately 70° in declination and by about 15° in inclination. These inclination data correspond to paleolatitudes of 12.4°N and 4.4°N for Ennery and Beloc, respectively.

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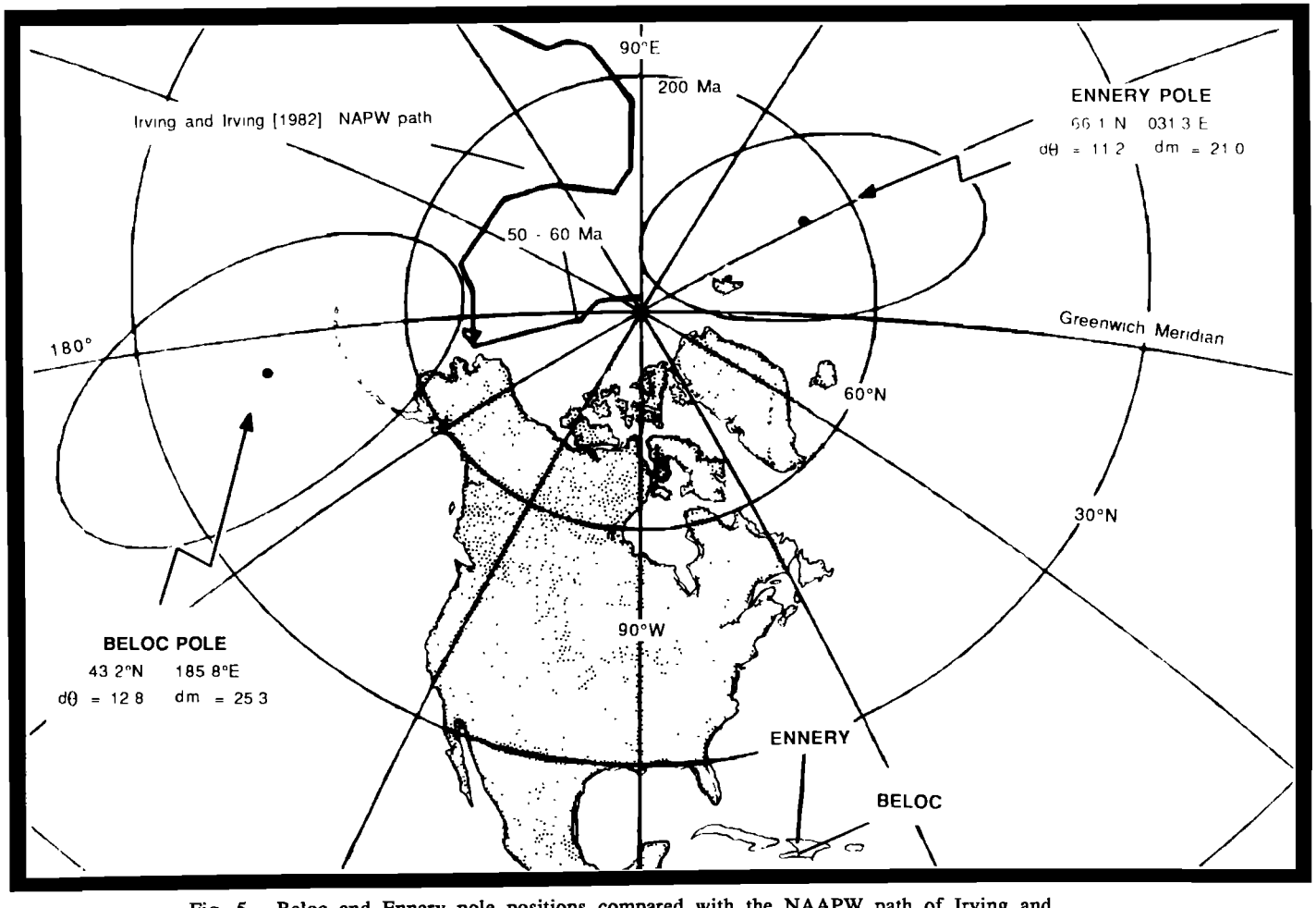


Fig. 5. Beloc and Ennery pole positions compared with the NAAPW path of Irving and Irving [1982]. Data for these poles are listed in Table 2. The Beloc pole is observed to be rotated anticlockwise relative to the reference pole (60 Ma) by about 33°. The Ennery pole is offset in a clockwise sense by about 37° relative to the 50 Ma reference pole. Each of these rotations is measured about a vertical axis situated at the corresponding sampling locality. Once rotated back toward the NAPW path, both poles remain "far-sided" (that is, offset southwest of the path), suggesting northerly translation of the sampling localities relative to North America since the Eocene.

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As shown in Figure 5, both poles are askant of the North American apparent polar wander path of Irving and Irving [1982]. Quantitative comparisons of Ennery and Beloc poles are made with the 50 and 60 Ma points of the path, respectively, as these ages correspond closely to the ages of the youngest rocks sampled at each locality. If we measure both of these offsets as angular rotations about vertical axes situated at the respective sampling localities, the Ennery pole ($\alpha 95 = 19.7°$) is rotated clockwise by 37° relative to the path, while the Beloc pole ($\alpha 95 = 25.1°$) is offset 33° in the anticlockwise sense. Discordance with the North American apparent polar wander path exists in terms of colatitude as well. If the poles are rotated about vertical axes back toward the path, they remain offset to the south (far-sided). This is an indication that both localities have suffered a northerly component of latitudinal displacement relative to North America since early Tertiary.

Both types of paleomagnetic disparity with respect to the North American apparent polar wander path are indicative of large displacements of the sampling localities relative to the North American plate. However, of equal importance are the differences that Ennery and Beloc locality data show relative to each other. Not only are the two locality means discordant by 70° in declination, but in terms of mean paleolatitude, the separation amounts to 8°. Today, the two localities are just 1° apart with respect to latitude (Figure 1).

We may consider the indicated displacements as tectonic in nature, i.e., they are not simply the result of local structural rotations of individual sites. A Caribbean plate tectonic model that treats the Ennery and Beloc localities as members of two segregated lithospheric blocks would be consistent with our findings. Two independent studies which invoke such a treatment have recently been proposed in the literature. Here we seek to integrate the paleomagnetics with appropriate aspects of these published tectonic models, with the hope of understanding our data in a plate tectonic context.

The reconstructions of Sykes et al. [1982] show significant latitudinal separation between northern and southern Hispaniola (Figure 6). In their model, emphasis is placed on northern Caribbean seismicity which yields a detailed account of plate boundary dynamics since the Late Miocene. Their analysis of seismic activity and geometry of downgoing seismic zones yielded a relative rate of motion of 3.7 cm/yr, which when extrapolated back to middle Eocene time, results in a total offset of 1400 km along the northern Caribbean plate boundary. This offset is the largest northern Caribbean displacement proposed in the literature, requiring complete closure of the Cayman trough at about 48 Ma.

At this time, northern Hispaniola, Cuba, Puerto Rico, and the Virgin Islands resided as a continuous island arc at the northern edge of the East Pacific-Caribbean plate. Sometime during perhaps the Oligocene, relative motion of the Caribbean plate with respect to the North American plate switches from northeasterly to easterly, giving rise to Lesser Antillean volcanism. As a result of this change in direction, the North American/Caribbean plate boundary shifts southward, transferring Cuba and northern Hispaniola to the North American plate. Predominantly left-lateral motion along this new plate boundary produces (or accommodates) seafloor spreading in the Cayman trough and obliquely subducts "proto-Caribbean" oceanic lithosphere. The Southern Peninsula of Haiti migrates northeasterly, and collision of the two Hispaniolan blocks is accommodated by horizontal crustal shortening sometime during the late Tertiary. The reconstructions of Sykes et al. [1982] suggest on the order of 5° latitudinal separation of northern and southern Hispaniola during the Eocene, as shown in Figure 6. Mean paleolatitude data from our study tend to support at least this aspect of the model.

Mann et al. [1984] have identified a large-scale, E-W oriented, left-lateral fault system which divides Hispaniola into two parts. This tectonic break, namely the Enriquillo-Plantain Garden Fault Zone, extends from the Cul-de-Sac plain in central Hispaniola westward along the Southern Peninsula to eastern Jamaica (see Figure 1). The fault is believed to extend farther west and merge with the Cayman Trough. It is interesting to note that across this fault zone the general strike of structural elements shows a similar relative angular offset as that revealed by the paleomagnetic data (see Figure 1). One such element is the Hispaniola Fault Zone, considered to represent early stage development of the Northern Caribbean Plate Boundary Zone [Mann et al., 1984].

By examining the distribution of recent (1950-1979) Caribbean earthquake epicenters, Sykes et al. [1982] noted a sublinear distribution of seismic activity in the vicinity of the Cul-de-Sac plain and Southern Peninsula. The activity is much less extensive than that along the present-day Northern Caribbean Plate Boundary Zone. However, as long as rates of relative motion along this northern left-lateral plate boundary exceed that of the left-lateral southern fault, then clockwise rotation of northern and central Hispaniola relative to both the Southern Peninsula and North America can occur. This sense of motion can account for the relative offset in structural trends and paleomagnetic data.

CONCLUSIONS

We have isolated stable remanent magnetizations in very weakly magnetized limestones and chalks from Haiti. The stable components can be characterized as having moderately high blocking temperatures, within-site bipolarities and two very different characteristic directions which are unique to two localities (Beloc and Ennery). We have shown that pole positions corresponding to Beloc and Ennery locality means are discordant with respect to

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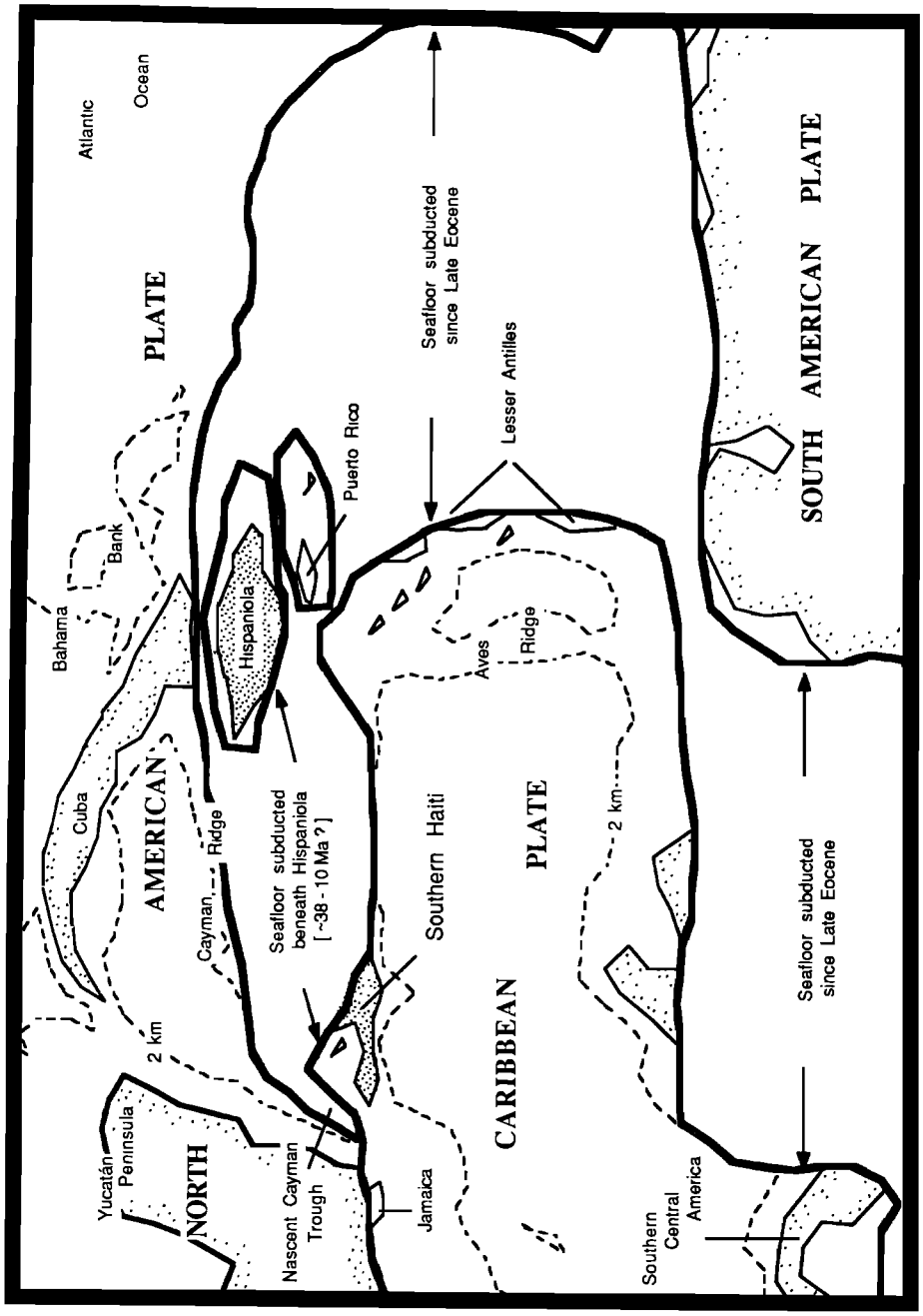


Fig. 6. Reconstruction of the Caribbean, North American, and South American plates in the Late Eocene (38 Ma) according to Sykes et al. [1982]. In this reconstruction the Cayman trough is completely closed, and a plate boundary exists between northern and southern Hispaniola. The Caribbean plate begins its easterly relative motion (1400 km) with respect to North and South America which results in subduction of "proto-Caribbean" oceanic crust. The latitudinal separation of northern and southern Hispaniola is supported by mean paleolatitude data from the Ennery and Beloc localities. (Figure redrafted from Sykes et al. [1982, Figure 11]).

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similarly aged North American reference poles. The tectonic-scale displacements we have described are non unique, simplified ways of accounting for the paleomagnetic discordance. However, the fact that comparable rotations and displacements have been suggested in the literature independently of paleomagnetism lends additional support to the rotations we have visualized.

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